

60-105 User Manual

LXI/USB 4-Slot Modular Chassis



LXI USB

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FM38792

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Product Safety

Safety Symbols

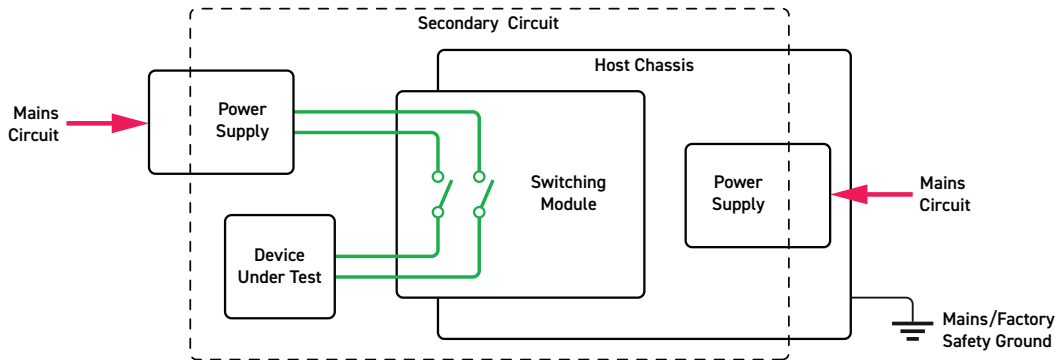
The following safety symbols may be used on the product and throughout the product documentation.

Meaning/Description	Symbol
PROTECTIVE EARTH (GROUND) To identify any terminal which is intended for connection to an external conductor for protection against electric shock in case of a fault, or the terminal of a protective earth (ground) electrode.	
DANGEROUS VOLTAGE To indicate hazards arising from dangerous voltages.	
WARNING/CAUTION An appropriate safety instruction should be followed or caution to a potential hazard exists. Refer to the relevant instructions detailed within the product manual.	
HEAVY If this product is heavy reference should be made to the safety instructions for provisions of lifting and moving.	
STATIC SENSITIVE To indicate that static sensitive devices are present and handling precautions should be followed.	

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Warnings & Cautions

WARNING - Hazardous Environments	Symbol
This product is not specifically designed for use in hazardous environments, for example in explosive atmospheres. If the product is to be used in hazardous environments we recommend that the user ensures suitable protective measures are taken.	
WARNING - Danger of Electric Shock	Symbol
This chassis may be populated with modules that contain hazardous voltages. Before removing the unit from the rack remove all supplies and disconnect user I/O signals. Unused slots in this chassis are populated with blanking panels to prevent access to user I/O signals that may be present. Blanking panels are available to order from Pickering. If the product is not used in this manner for example by using an extender card then additional care must be taken to avoid contact with exposed signals. Not to be used in safety critical circuits, refer to the Pickering Interfaces' terms & conditions of sale. Installed modules must not be used for the switching of Mains Circuits. For the switching of voltages up to the unit's full specification, Secondary Circuit power supplies and the Device Under Test must be fully isolated from the Mains Circuit and the safety earth. The chassis must be correctly installed and be connected to mains or factory safety ground, please refer to the installation section of this manual.  Diagram Showing Separation Between The Mains Circuit and Secondary Circuit / DUT For the purposes of this document, a Mains Circuit is an installation classified as Overvoltage Category II up to 300 V. Separation between the Secondary Circuit and Mains Circuit is achieved by a transformer in which the primary windings are separated from the secondary windings by reinforced insulation, double insulation, or a screen connected to the protective conductor terminal.	
WARNING - Equipment Grounding	Symbol
This module is powered by a class I external power supply which must be connected to the mains supply and electrical ground by the supplied three core IEC cable. The module chassis is also connected to electrical ground via the DC connector from the power supply.	

Warnings & Cautions

CAUTION - Handling of Electrostatic-Sensitive Devices	Symbol
Certain semiconductor devices used in this equipment are liable to damage due to static voltage. Observe the following precautions when handling these devices in their unterminated state, or sub-assemblies containing these devices: • Persons removing sub-assemblies from equipment using these devices must be earthed by a wrist strap and a resistor at the point provided on the equipment. • Soldering irons used during the repair operations must be low voltage types with earthed tips and isolated from the mains voltage by a double insulated transformer. • Outer clothing worn must be unable to generate static charges. • Printed Circuit Boards (PCBs) fitted with these devices must be stored and transported in anti-static bags.	

CAUTION - Product Documentation	Symbol
Suitably qualified & trained users should ensure that the accompanying documentation is fully read and understood before attempting to install or operate the product.	

Safety Instructions

Safety Instructions
All cleaning and servicing requires the equipment to be isolated and disconnected from the power source and user I/O signals (refer to the Maintenance Section). • Appropriate manual handling procedures should be followed as dictated by the weight of the individual module or the combined weight of the modules & chassis. • Should a fault occur with the module or chassis, immediately isolate and disconnect the incoming power to the chassis and the user I/O signals. • Ensure the equipment is installed, operated and maintained by trained and authorised personnel. • For suitably equipped products in the event of an emergency press the red "emergency stop" button situated on the front of the unit.

Section 1 - Technical Specification

- Fully Compliant LXI Interface (1.5)
- Ethernet 1000baseT Interface
- USB 3 Compatible
- Supports 1000+ Pickering PXI 3U Modules Including:
 - Programmable Resistors
 - Matrices
 - Multiplexers
 - General Purpose Relays
 - RF Switches
 - Fault Insertion
- Supports Four User Slots
- Compact 1/2 Rack Width 2U Form Factor
- 3 Year Warranty



Pickering Interfaces' 60-105 modular chassis offers a small lightweight form factor, making it ideal for portable, benchtop, and space restrictive applications.

The 60-105 supports between one and four 3U PXI modules and accepts Pickering's PXI products, it is the perfect platform to construct small scale applications.

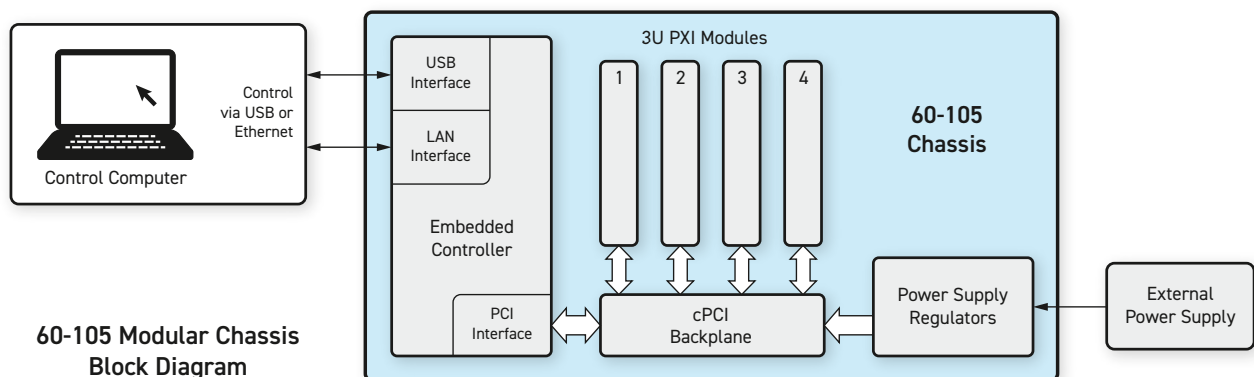
Possible systems include switching matrices up to 3072 crosspoints (one 40-558 BRIC4 occupying all four slots) or up to 72 channels of programmable resistor/sensor emulation (four 40-295 18-channel resistor modules occupying four slots).

For a list of compatible PXI modules, please refer to the 60-105 user manual.

The USB compatible and LXI compliant interfaces enable the chassis to be controlled directly through standard interfaces found on most personal computers, allowing for a very cost effective route into the modular test and measurement market.

The chassis can be controlled through built in software drivers. Chassis configuration and module control is possible through the W3C web page interface.

The 60-105 is powered from an external DC supply, making it lightweight, versatile and extremely portable.



60-105 Modular Chassis
Block Diagram

Section 1 - Technical Specification

Specifications

Chassis Backplane:	32-bit cPCI backplane, compatible with PXI.
Chassis Capacity:	4 off 3U user slots available.

PXI Module compatibility

The chassis is supplied with drivers for Pickering PXI modules.	
Switching Support:	Most of 3U Pickering Interface's PXI switching modules. Includes (but not limited to): All 2 & 4-slot 3U BRIC matrices, featuring up to 2208 crosspoints.
Simulation support:	All programmable resistor and potentiometer modules offering up to 72 channels, RF attenuators, digital I/O and other simulation modules.

Power Supply

Input Voltage Range:	Use the supplied PSU or an external supply of 19 V $\pm$ 5 %
Input Current Rating:	5.2 A (DC)

Cooling

Airflow:	Crossflow
Per-slot Cooling Capacity:	25 W at 40 °C ambient
Fans:	1 off 7.7 cfm fan and 1 off 33 cfm fan

Acoustic Noise Emissions

With fans on maximum:	53 dBA typical
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LAN Interface

Designed to comply with the LXI Standard Version 1.5	
Connector:	RJ45 Connector.
Connection Speed:	1000BaseT interface.

USB Interface

Compatible with USB3 (backwardly compatible with USB/USB2)	
Connector:	USB3 type B
Connection Speed:	400 MBps

Mechanical Specification

Dimensions:	
With Rubber Bumpers:	Width: 215.5 mm (8.48") Height: 98.1 mm (3.86") Depth: 311 mm (12.24")
Without Rubber Bumpers:	Width: 201.1 mm (7.92") Height: 85 mm (3.35") Depth: 301 mm (11.85")
Weight:	3.0 kg without PXI modules
External PSU Dimensions:	W: 58 mm, H: 37 mm, D: 142 mm
Lead Length:	1200 mm

Slot Capacity (Per Slot)

+3.3 V	+5 V	+12 V	-12 V
6 A	6 A	1A	1A

Note: Maximum power is 60 W for all slots or 30 W per slot.

Operating/Storage Conditions

Operating Temperature:	0 °C to +55 °C
Humidity:	Up to 90% non-condensing
Altitude:	5000 m
Storage/Transport Temperature:	-20 °C to +75 °C
Humidity:	Up to 90% non-condensing
Altitude:	15000 m

Safety, CE & RoHS Compliance

All products are fully CE compliant and meet applicable EU directives:

Low-voltage safety EN61010-1:2010,
EMC Immunity EN61326-1:2013,
Emissions EN55011:2009+A1:2010.

The 60-105 Chassis also complies with the European Restriction of Hazardous Substances directive (RoHS).

Included in the Package

The 60-105 chassis is shipped as a package which includes:

- Pickering LXI chassis (60-105-001) with blank panels fitted to all slots
- USB3 Cable and Ethernet Cable
- 19V Power Supply and appropriate power cord for your territory
- Product Information/Driver DVD

Section 1 - Technical Specification

Product Order Codes

LXI/USB Modular Switching Chassis, 4-Slot	60-105-001
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Accessories

Replacement 19V DC Power Supply	63-104-001
Optional Chassis Strap Mounting Kit*	63-105A-003
Optional Rack Shelf for 1 or 2 off 60-105 Chassis*	63-105-004
Optional Chassis Low Profile Plate Mounting Kit*	63-105-005
Optional Full Shelf Mount	63-105-007
PXI 1-Slot Blanking Kit	40-942-001
PXI 1-Slot Air Blocker	40-942-011

* Please refer to the 60-105 User Manual for information

The 60-105 is part of a range of PXI and PXIe Hybrid chassis which includes PXI chassis with an LXI or USB control interface.

Pickering's Range of PXI & PXIe Chassis			
Chassis Type	Number of Slots	Chassis Size	Model No.
PXI	14	4U Full Rack	40-914
PXI	19	4U Full Rack	40-923A
PXI	8	4U Half Rack	40-924
PXIe Hybrid (Gen 3)	8	4U Half Rack	42-924
PXIe Hybrid (Gen 3)	18	4U Full Rack	42-925
PXIe Hybrid (Gen 2)	18	4U Full Rack	42-926
PXIe Hybrid (Gen 2)	21	4U Full Rack	42-927
PXI with LXI/USB Control	7	4U Half Rack	60-102D
PXI with LXI/USB Control	18	4U Full Rack	60-103D
PXI with LXI/USB Control	2	1U Half Rack	60-104
PXI with LXI/USB Control	4	2U Half Rack	60-105
PXI with LXI/USB Control	6	1U Full Rack	60-106



Section 2 - Technical Description

Mechanical Description

The 60-105 Chassis comprises the following:

- 216 mm x 98 mm x 311 mm chassis suitable for desk top operation.
- 4-slot cPCI backplane, PXI compatible. Does not implement PXI Trigger Bus or 10 MHz clock.
- Built-in microcontroller providing the interface between the backplane and the control ports.
- External 19 volt power supply providing power for the unit.
- Power regulators providing power for the microcontroller, backplane and PXI cards.
- Front panel mounted status LCD Display.
- Rear panel mounted status LEDs.
- Rear panel mounted RJ45 1000baseT LAN connection for LXI control of the unit over Ethernet.
- Rear panel mounted USB type B connection for control of the unit over USB.
- Rear panel mounted power plug for connection to the external power supply.



Section 2 - Technical Description

Functional Description

The 60-105 LXI/USB Chassis is capable of supporting a wide range of Pickering PXI switching modules in a self-contained unit, the only connections required are power and an Ethernet or USB lead. The chassis consists of a 4-slot cPCI backplane (compatible with PXI) and a controller assembly with power being supplied from internal regulators which are driven from an external mains power unit. Control and programming information for the PXI modules installed in the chassis is sent to the backplane via a PCI bridge device on the controller assembly. Programming information for the controller is sent to the chassis via the USB interface or the LAN interface from a remote computer using the chassis' unique Ethernet address. Programming documentation for the chassis as well as data for Pickering's compatible switching products is stored in the embedded processor on the controller assembly and can be downloaded via the USB or LAN interface. The LCD display on the front panel provides an indication of the unit's status and LEDs on the rear of the unit indicate condition of the LAN interface.

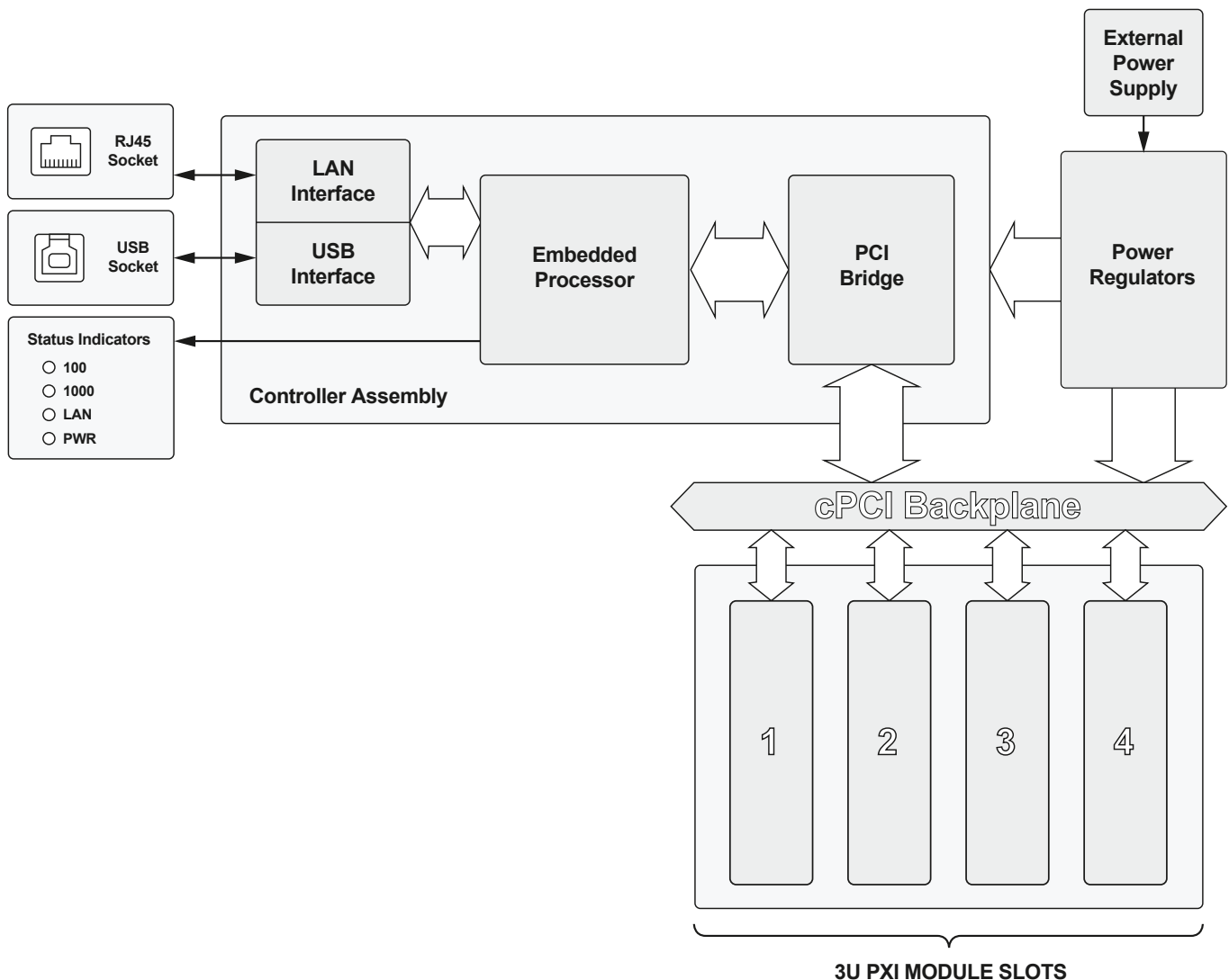


Figure 2.1 - Block Diagram for the 60-105 4-Slot LXI/USB Chassis

Section 2 - Technical Description

PXI Module Support

The 60-105 supports a wide variety of Pickering Interfaces 3U PXI modules, the backplane of the chassis is conformant to the PXI standard. The chassis' LXI interface is conformant with the stated version of the LXI standard.

All Pickering Interfaces' 3U single, dual, triple and 4-slot slot PXI switching modules and resistor modules are supported with the exception of the following:

- 41-610 - Arbitrary Waveform Generator
- 41-620 - Function Generator
- 41-740 - Programmable Power Supply
- 40-738 - USB 2.0 HUB With Programmable Connect/Disconnect

No third party PXI modules are supported.

The PXI modules are provided with a variety of drivers and this includes an IVI driver as required by the LXI standard, the drivers supported are described in the software section of this manual. Where PXI modules state an operating speed the specification refers to the speed when supported in a PXI chassis which is lightly loaded and not subject to operating system delays, the latency in an Ethernet connected system will increase the operate time under the same conditions (for more information consult you local Pickering Interfaces sales office).

Where speed is important to an application we generally recommend that the IVI driver is not used since IVI imposes additional delays and prevents more efficient operation of the switches by bulk changes (adding more than one path connection in a single operation) – the IVI software model has no facility for this function.

Section 3 - Installation

Refer to the Warnings and Cautions at the front of this manual



The chassis is designed for indoor use only.

Installing the LXI Chassis

Site Considerations

The 60-105 is designed to operate on a bench or desk top and should be positioned horizontally so the cooling air vents are not restricted.

Cooling Considerations

The unit is cooled by internal fans which draw air via two intakes in the side of the chassis and expel air through two vents on the rear panel. Cooling air for the PXI modules enters through the larger of the two intakes in the side of the chassis, and is expelled via the larger vent in the rear panel. To maintain the correct internal air flow, ensure blanking panels are fitted to any of the PXI slots if they are unused. The controller and power regulator area of the chassis is cooled by air drawn through the smaller side intake and is expelled via the smaller vent on the rear. It is important for the correct functioning of the unit that all cooling air inlets and outlets are unrestricted and objects such as cabling are kept clear.

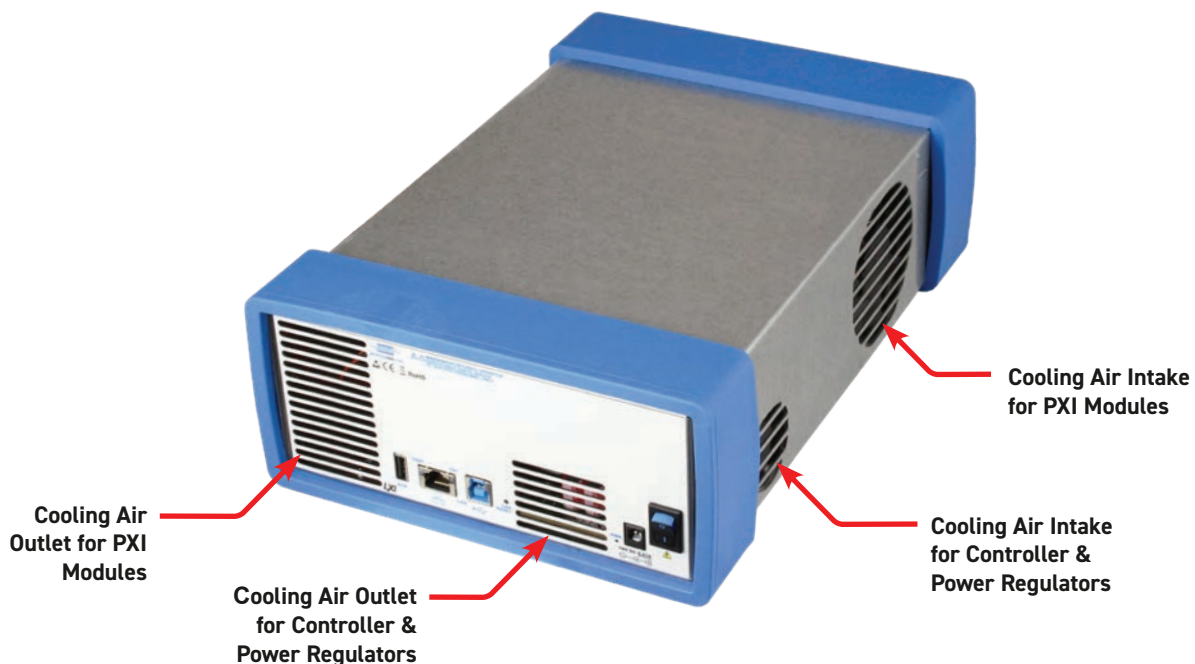


Figure 3.1 - Rear View of 60-105 LXI/USB Chassis Showing Cooling Vents

Section 3 - Installation

Connecting the Chassis

The only connections required to use the 60-105 Chassis are power and a control cable. Power is supplied from an external power unit supplied with the chassis, and control can be via the USB3 type-B socket or the RJ45 Ethernet port on the rear panel. The positions of these connectors are shown in the figure below:

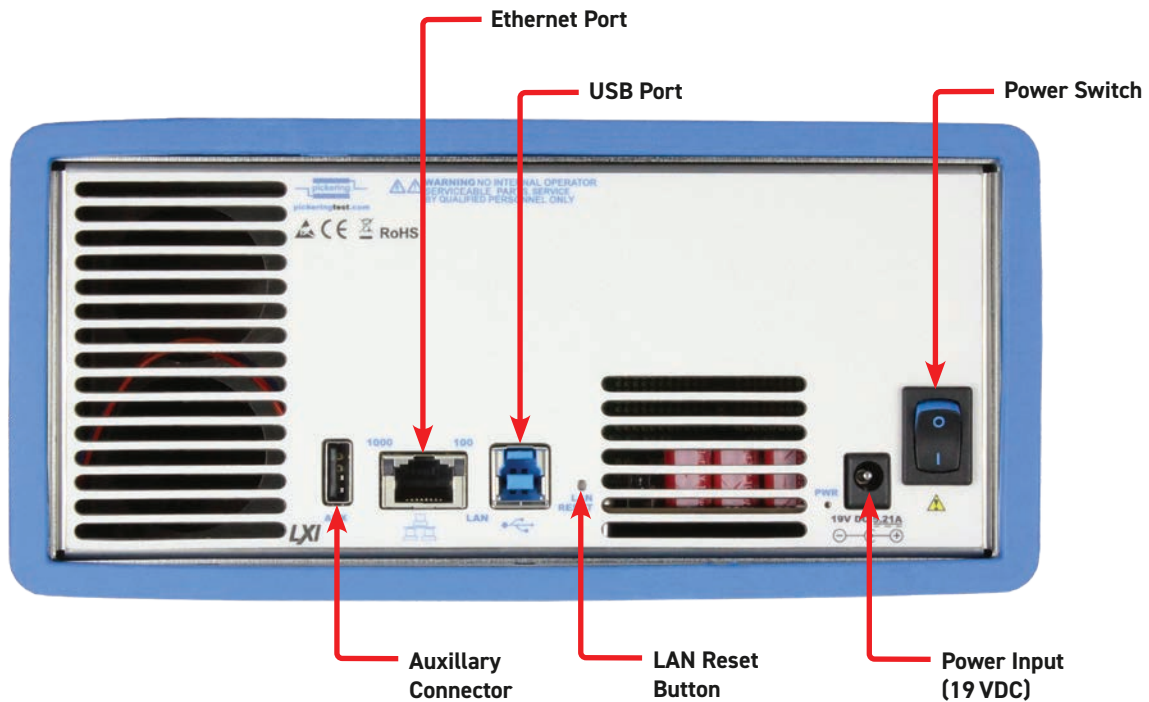


Figure 3.2 - Rear Panel of 60-105 Chassis Showing Positions of Connectors and Switches

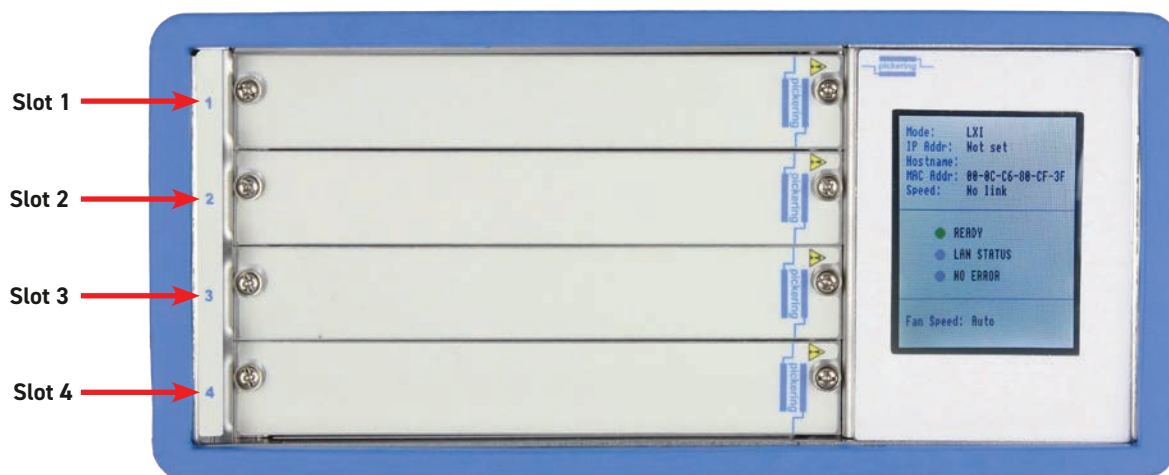


Figure 3.3 - Front Panel of 60-105 Chassis Showing PXI Slot Positions

Section 3 - Installation

Installation of Individual Modules

Modules should be installed in accordance with the following procedure:



1. Ensure that the chassis is turned OFF.
2. Choose an appropriate PXI slot in the chassis.
3. Remove the blanking plate for the chosen slot.
4. Ensure that the injector/ejector handle is in its downward position. Align the module with the card guides on the top and bottom of the slot.
Do not raise the injector/ejector handle whilst inserting the module as the module will not insert properly.
5. Hold the handle whilst slowly sliding the module into the card guides until the handle catches on the injector/ejector rail (refer to Figure 3.4).
6. Raise the injector/ejector handle until the module firmly seats into the backplane. The front panel of the module should be flush with the panel in the adjacent slot and fully seated against the chassis rails.
7. Secure the front panel of the module using both screws to the chassis mounting rails.

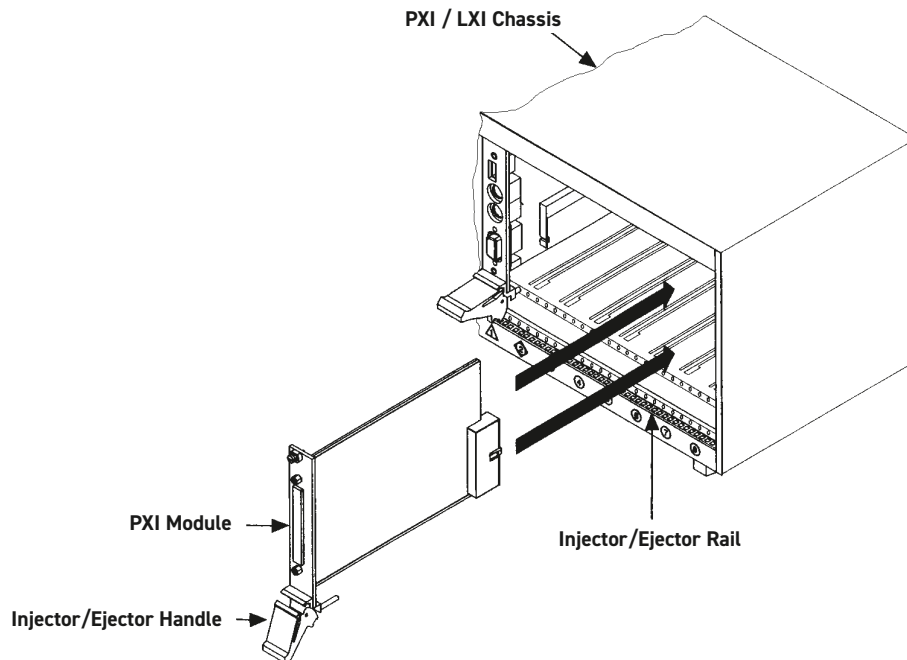


Figure 3.4 - Example Installation of a PXI Module into a PXI/cPCI/LXI Chassis

Section 4 - Configuration & Set-Up

Default Configuration

The 60-105 Modular Switching Chassis has factory default settings causing it to behave in the ways listed below. If the user makes changes to the chassis' configuration pages this behaviour may no longer apply.

1. Ethernet cable only connected:

Operation is as a standard LXI product.

2. USB Cable only connected:

Operation is no longer LXI, connection is made to the host computer through the RNDIS protocol and will appear as a network adapter on that machine.

3. USB Cable connected as well as an Ethernet cable:

The USB connection is always the priority connection so LXI mode is disabled. Fundamentally the same as configuration 2. The Ethernet interface is not configured.

Changes can be made to the chassis' USB configuration page to set the secondary interface to be the Ethernet port. This secondary interface is not LXI compliant and is for the purposes of monitoring software. If an IP Address is set, the interface will broadcast router messages and is able to accept SSH terminal sessions.

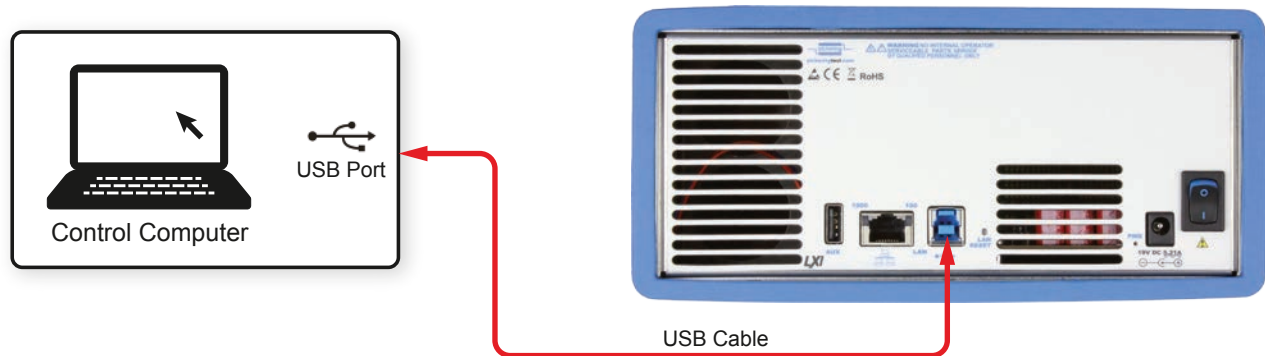
A reset using the recessed 'LAN Reset' button on the rear panel of the chassis will work regardless of the cable connection status of the unit and will cause the chassis to revert to the factory default configuration.

Section 4 - Configuration & Set-Up

USB Configuration

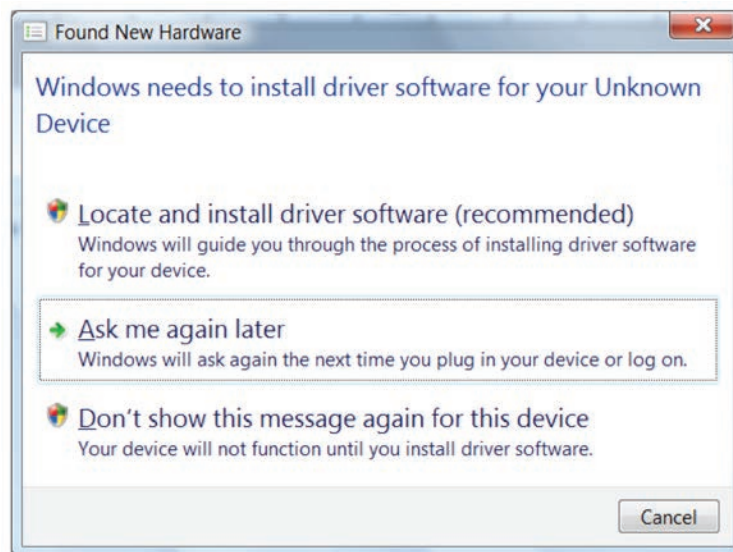
To use the 60-105 Modular Switching Chassis over a USB connection it should be set up and configured using the following instructions:

1. Insert your chosen PXI cards into the chassis as per the instructions in Section 3.
2. With the chassis powered off, insert the supplied USB3 cable into the Type B socket on the rear of the chassis marked with the  USB symbol.



3. Insert the other end of the cable into a free USB port on your host computer, then power on the 60-105 chassis.

If this is the first time the chassis has been connected to your computer, a dialog screen will be displayed:



4. Select **Locate and install driver software**. You will then be asked to insert the supplied 60-105 driver DVD.
5. Once the Found New Hardware wizard has completed a new device will be installed under network adapters listed as:
Pickering 60-105-001
6. Complete the installation by running the installation package **ClientBridge** located on the supplied DVD.

Section 4 - Configuration & Set-Up

LXI LAN Configuration

To use the 60-105 LXI Modular Switching Chassis over a LAN connection it should be set up and configured as described in the "LXI Getting Started Guide" document. The device's home page is shown below:



Figure 4.1 - Modular Chassis Instrument Home Page

If the IP address is to be set manually and before connecting to a network, contact your IT department so they can give you the correct settings for:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server(s)
- Domain name

Once you have the required information, the IP configuration window is used to manually set the IP address. This is shown in the figure below. Incorrect setting will result in a loss of communication indicated by a red LAN LED on the rear panel of the unit. If this does occur press the LAN Reset on the rear panel for a full factory reset.

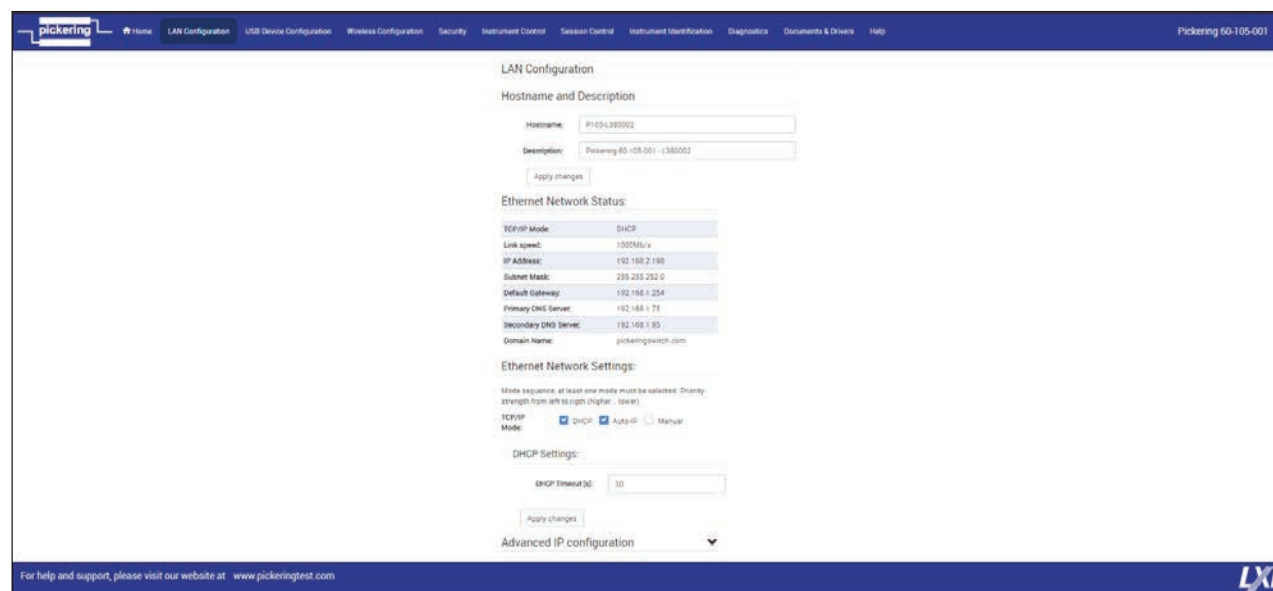


Figure 4.2 - Modular Chassis LAN Configuration Page

Once correctly set, the current IP address of the chassis is indicated on the LCD display on the front panel. The location of this is shown in Figure 4.4 overleaf.

Section 4 - Configuration & Set-Up

Chassis Status Indicators

The 60-105 chassis includes indicator LEDs on the rear panel as well as indicators on the front panel display to show the status of the chassis and the LAN interface. The positions of these indicators are shown in the figures below and their meaning is outlined in Table 4.1.

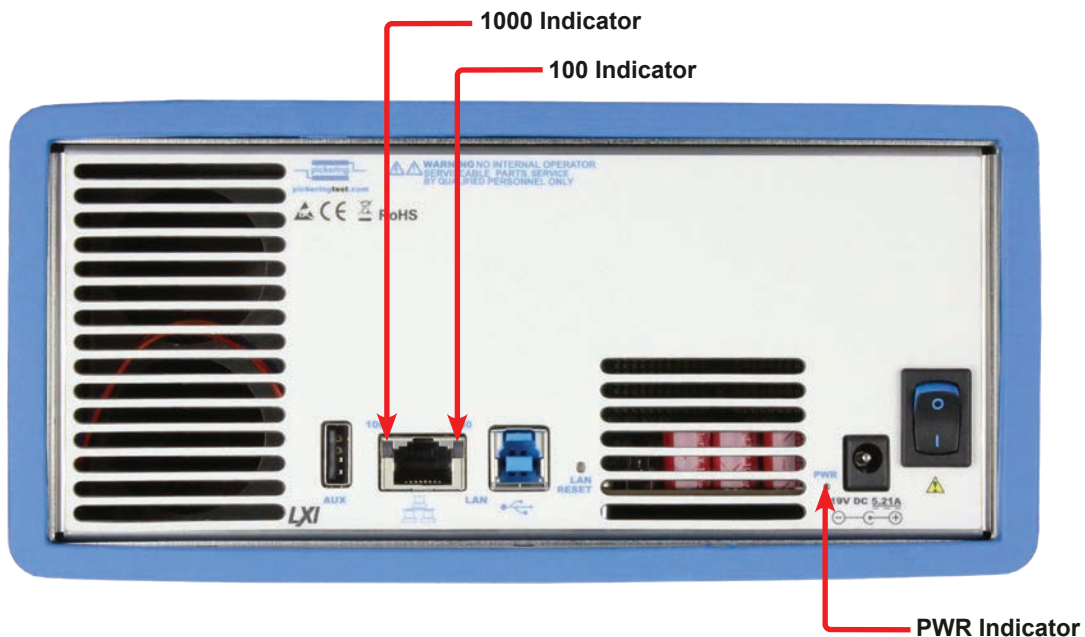


Figure 4.3 - Rear Panel of 60-105 Chassis Showing Status Indicator Positions

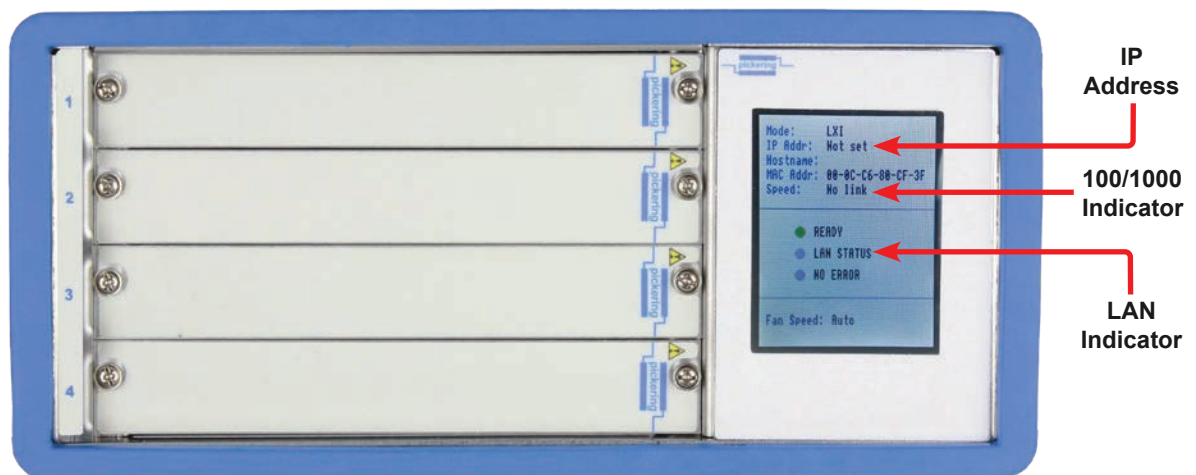


Figure 4.4 - Front Panel of 60-105 Chassis Showing LAN Indicators and Current IP Address

Section 4 - Configuration & Set-Up

Table 4.1 - Status Indicator Meaning

Indicator Name	Indicator Meaning		
	On - Steady	On - Flashing	Off
PWR	Green - Power On	-	Power Off
LAN	Green - Ethernet On Red - System Error	Green - LXI Identification Red - IP Conflict	USB Mode
1000	1000BaseT	-	Both off with "LAN" on - 10BaseT
100	100BaseT	-	

Section 5 - Programming

Programming Options for Pickering Interfaces LXI Units

For information on the installation and use of drivers and the programming of Pickering's products in various software environments, please refer to the Software User Manual.

This is available as a download from:

pickeringtest.com/support/software-drivers-and-downloads

Also, it can be accessed from the DVD supplied with the product:

`<DVD_DRIVE>\downloads\Documentation\Software-Manual.pdf`

Section 5 - Programming

Unit Architecture

Before discussing individual drivers it is important to understand the basic architecture of Pickering LXI units.

IP Address, Bus and Device

Each LXI unit has a unique IP address on your network. This address will depend on how you have configured your LXI unit and on network configuration. As delivered most LXI units are configured for dynamic address allocation, that is your network is expected to provide the IP address for the unit as part of its power-up sequence. This address may be configured as a fixed address, but there are possible problems, refer to the Getting Started guide for details on how to discover and configure your LXI unit.

Within an LXI unit, each switch card occupies a PCI slot, and so has a unique bus/device location with the embedded computer. It is vital to know that location in order to specify the address of the switch to be controlled.

Pickering provide software tools to help you discover the bus and device number of the cards in your LXI unit. Refer to the distribution disk that came with your unit in the folder LXI Products/Tools.

Sub-units

Pickering PXI cards contain one or more independently addressable functional blocks, or sub-units. Sub-unit numbers begin at 1, and separate sequences are used for input and output functions. This number is used in function calls to access the appropriate block. Generally, sub-unit numbers correspond directly to the bank numbers specified in hardware documentation.

Sub-unit examples:

Model	Configuration	INPUT sub-unit #1	OUTPUT sub-unit #1	OUTPUT sub-unit #2	OUTPUT sub-unit #3
40-110-021	16 SPDT switches	None	16 SPDT switches	None	None
40-290-121	Dual Programmable resistors + 16 SPDT switches	None	Resistor #1	Resistor #2	16 SPDT switches
40-490-001	Digital I/O	16-channel inputs	32-channel outputs	None	None
40-511-021	Dual 12 x 4 matrix	None	12 x 4 matrix #1	12 x 4 matrix #2	None

Sub-unit characteristics

The numbers of input and output sub-units in a card can be obtained using functions from the library:

For example, in the basic piplx library:

```
PIPLX_EnumerateSubs  
PIPLX_SubType, PIPLX_SubInfo
```

Are used to identify the individual sub-units on a card.

Section 5 - Programming

Controlling a Switch

To control a Pickering switch the command must contain the full location of the sub-unit being controlled; the IP address of the LXI unit, the bus/device location of the particular card in that LXI unit, and the sub-unit which contains the switch to be controlled.

A number of tools are available to locate the IP address of an LXI unit. Refer to the "Getting Started" guide delivered with your unit, also available on the distribution disk in the "LXI Products" folder. Only Pickering tools such as "Find LXI" will identify the switch card bus/device location. Pickering Direct I/O driver includes functions to allow a Pickering LXI unit to be found programmatically.

To control a card there are a number of steps required. For example using PIPLX it is necessary to first open a session on a specified LXI unit, then to obtain a handle to a particular card in the LXI unit and finally to issue commands to a particular sub-unit on that card.

For example, to access the card at bus 0, device 20, on the LXI unit at 192.168.1.100 :

```
PICMLX_Connect(0, "192.168.1.100", 1024, 2000, &session)
PIPLX_OpenSpecifiedCard(session, 0, 20, &card_num)
PIPLX_OpBit(session, card_num, sub-unit, bit, state)
```

Using the IVI driver, the connect and open functions are combined in the `pi40iv_InitWithOptions` function where the VISA resource string will specify the IP address and the card bus/device location, for example:

```
pi40iv_InitWithOptions("TCPIP::192.168.1.100::0.20::INSTR",...)
```

Here the resource string identifies an IP address of 192.168.1.100 and then the bus, device location as 0.20, that is bus 0, device 20.

For full details of how to access and use Pickering drivers the user is advised to refer to the individual documentation packs provided with the installed drivers, this can be found on the distribution disk that came with your unit.

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Installing Drivers

Please download and Install ClientBridge Driver from the DVD and Website.

Website:

pickeringtest.info/downloads/drivers/Sys60/

DVD:

<PATH_TO_DVD_DRIVE>\Drivers_LXI\

The ClientBridge Driver comes with C/C++ and .NET libraries and also provides various example programs and user applications to get the user up to the speed.

USB Drivers

Pickering provides digitally signed USB driver as part of ClientBridge. Once the drivers are installed, the USB chassis will appear under **Network adapters**.

Note: Each subsequent USB port used on a specific PC will result in a unique identifier being assigned under Network adapters. Figure 5.1a illustrates the identifier for the first installation; Figure 5.1b shows the identifier for the second USB port; Figure 5.1c the third USB port installation.

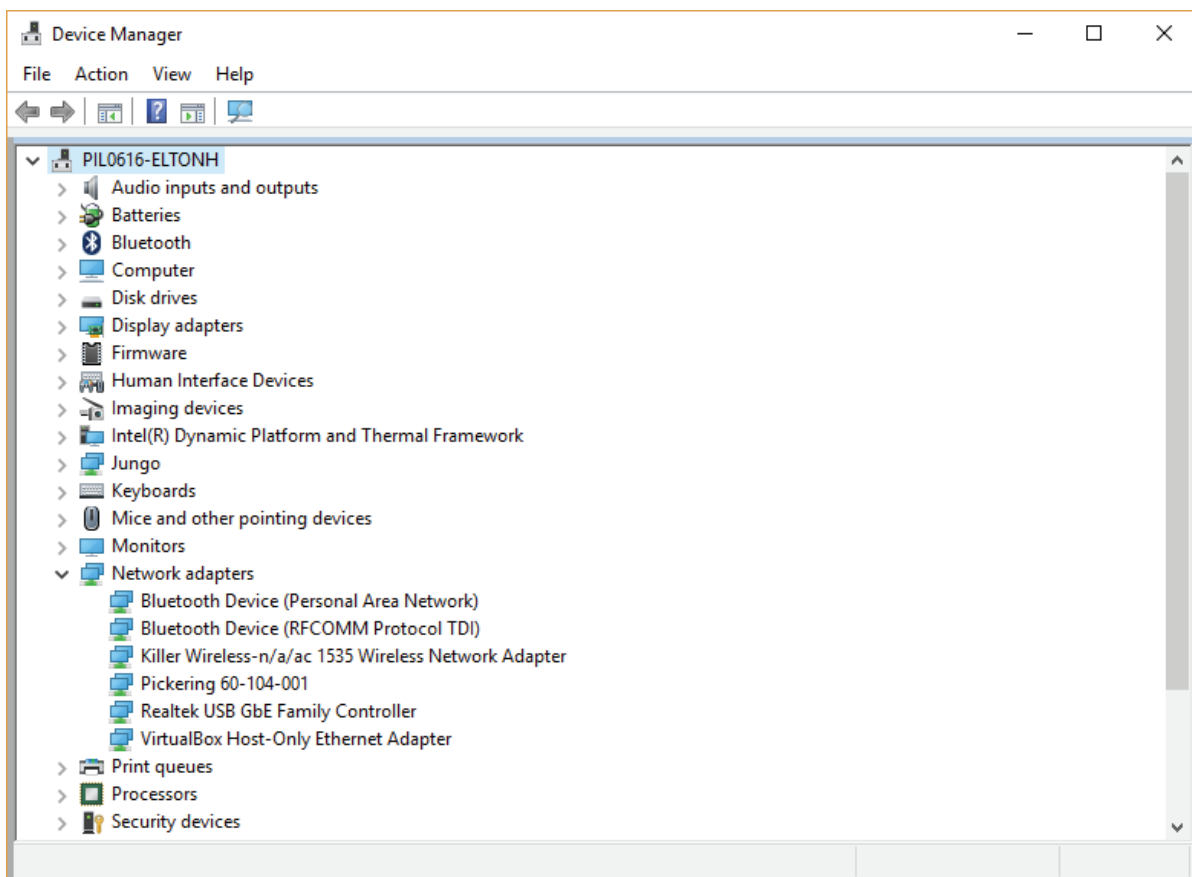


Figure 5.1a - Similar to above, the 60-105 Chassis will appear in the List of Network Adapters

Section 5 - Programming

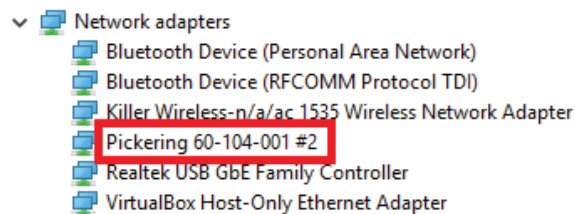


Figure 5.1b - Similar to above, the 60-105 Chassis is Connected to a Second USB Port

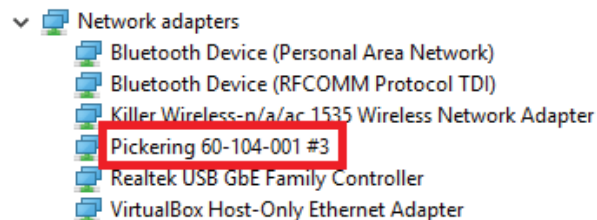


Figure 5.1c - Similar to above, the 60-105 Chassis is Connected to a Third USB Port

LXI Discovery Tool

The LXI Discovery Tool uses different discovery protocols to locate LXI Devices on your network, the place where your computer and LXI Device can communicate with each other. The device can be discovered irrespective of USB, Wi-Fi or LAN interface.



Figure 5.2 - LXI Discovery Tool Window

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General Soft Front Panel

Pickering provides General Soft Front Panels for all our PXI/PCI and LXI products to get the user familiarize with our products and get up to speed. Similar to our drivers, our Soft front panels cover all of our cards and dedicated views are provided for selected products. So whether you buy a Fault Insertion Card, a Resistor Card or a Matrix, all our products can be operated from the same SFP.

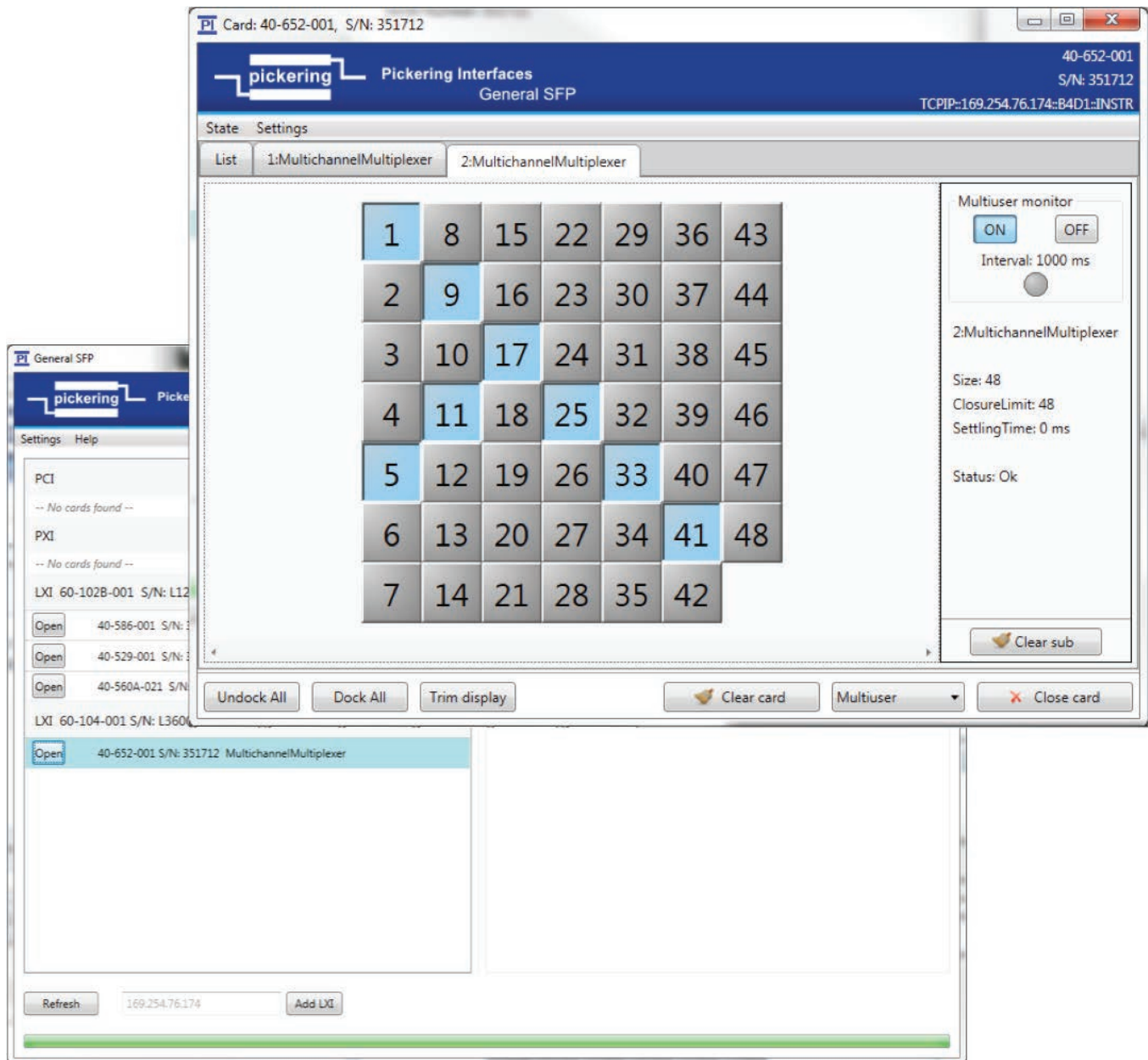


Figure 5.3 - Example of a General Soft Front Panel

Section 5 - Programming

HTML5 Soft Front Panel

The HTML5 Soft Front Panel provides platform independent control of Pickering products allowing their use from various PCs (including Linux OS), tablets and other mobile devices. The user can access the HTML5 Soft Front Panel by navigating to the LXI home page and clicking the **Instrument Control** tab.

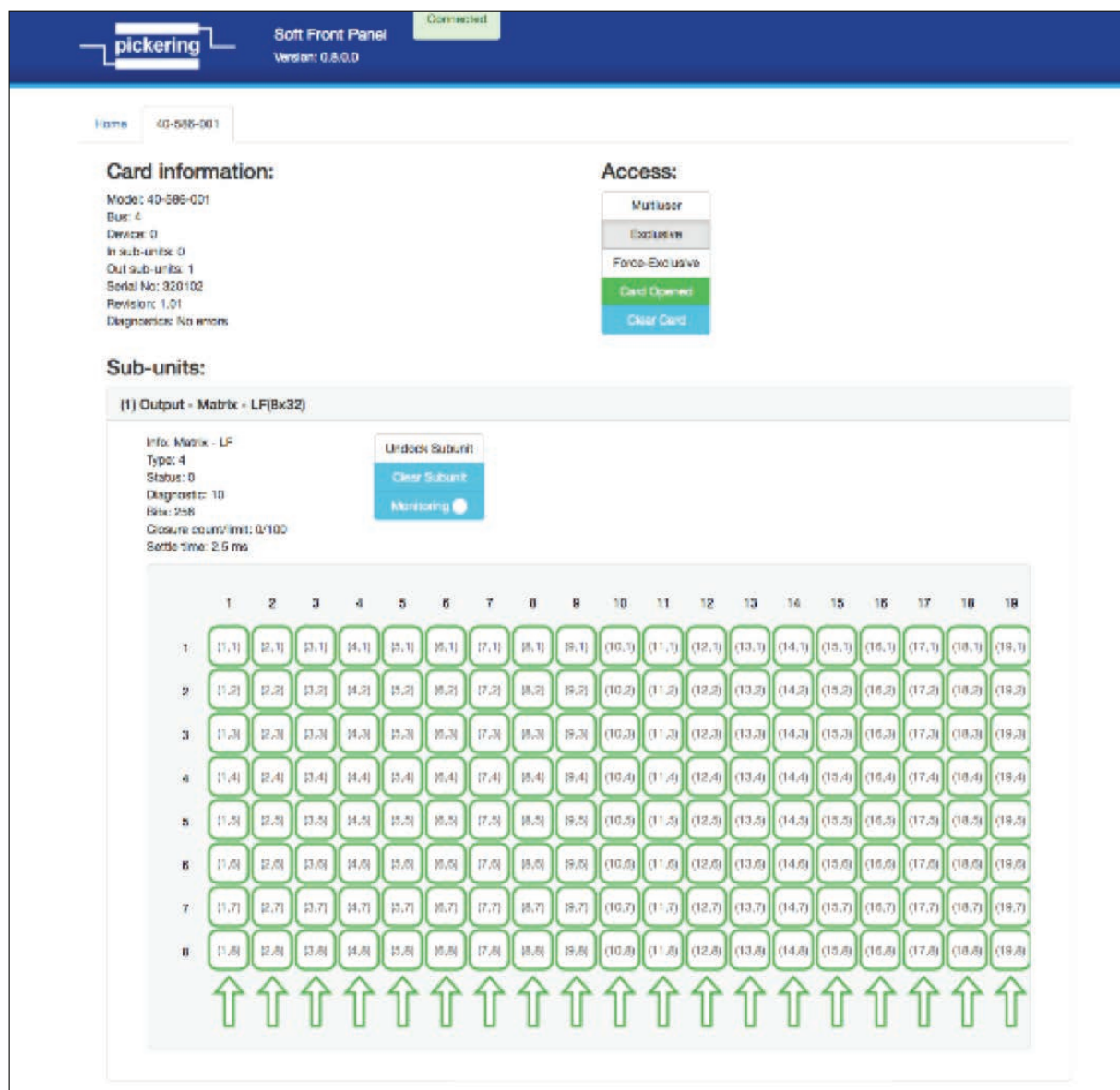


Figure 5.4 - Example of an HTML5 Soft Front Panel

Section 5 - Programming

Resource Manager

Pickering's LXI and PXI products can be controlled and operated via aliases using the Pickering Resource Management Library. It allows the user to define aliases for every Pickering PXI and LXI module when writing a test application. This is useful because any change in the hardware interfaces will not affect the test application and only requires modification of the configuration file to point to the right alias.

You can use the Pickering Resource Manager application to define your aliases for the available devices. Alternatively you can use C++ API (Pirml.dll library) and .NET C# connector (Pickering.Resources.PirmlInterop.dll). For more information see the Language Support topic.

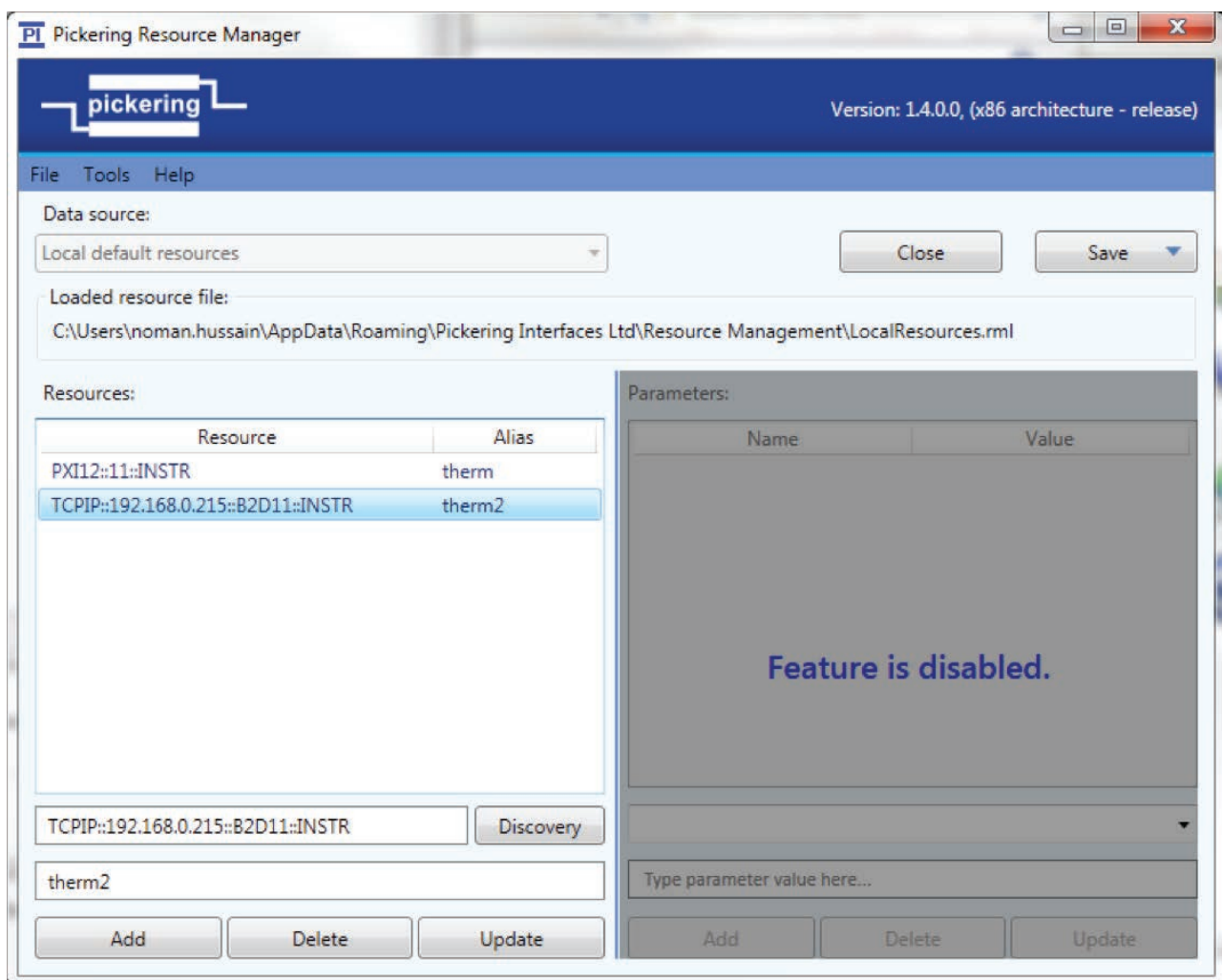


Figure 5.5 - Pickering Resource Manager Window

Section 5 - Programming

Command Line Control of Pickering LXI Instruments

Overview

Command line control of the Pickering LXI device range was added in firmware release 2.11. The utility provided was ported to the LXI instrument from the Windows utility PILMon and provides identical functionality.

Enabling the Command Line Control Utility

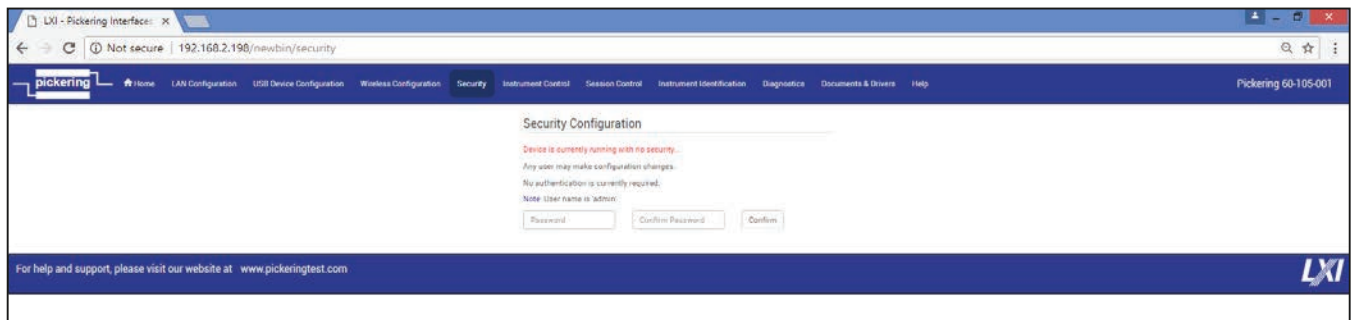
Pickering LXI devices do not provide an SSH server in the normal configuration state, the service must be explicitly enabled.

To do this the user must first enable security, this helps to prevent inadvertent access by limiting the accessibility of instrument services, principally by requiring password access to control functions. Once security is enabled, the SSH server may be started.

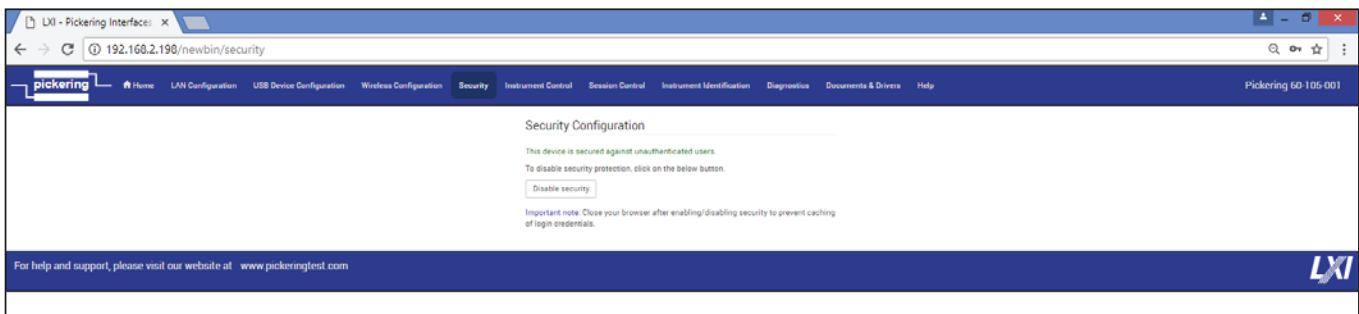
All the following assumes that the user knows the IP address of the LXI device. If this is not known, a number of tools are available, many on the Pickering distribution disk, to allow the IP address to be discovered.

Step 1 - Enable Security

Select the Security Page from the main menu list



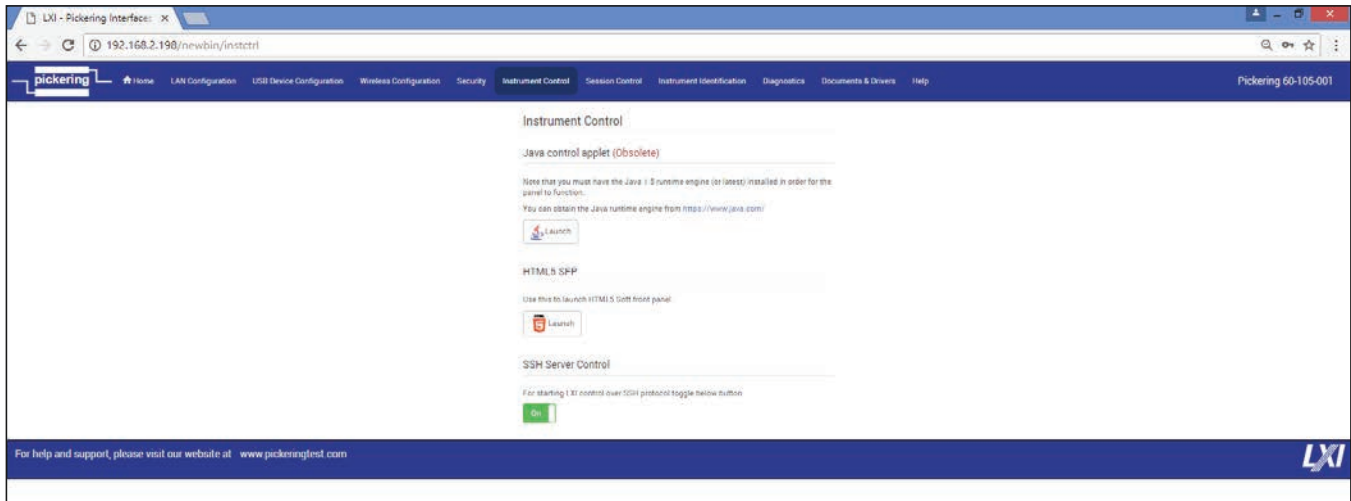
Enter a password and confirm then click on the Submit button.



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Step 2 - Enable the SSH server

Select Instrument Control from the main menu



NOTES:

The password and state of the SSH server are stored in non-volatile memory within the LXI device and so will remain in effect even after a power cycle. It is important to remember the password applied. Should the password be forgotten, the device may be reset to initial conditions by operating the LAN Reset button located on the front of the controller module in the appropriate manner, refer to Section 6 of this manual for details of this process.

Disabling security will stop the SSH service.

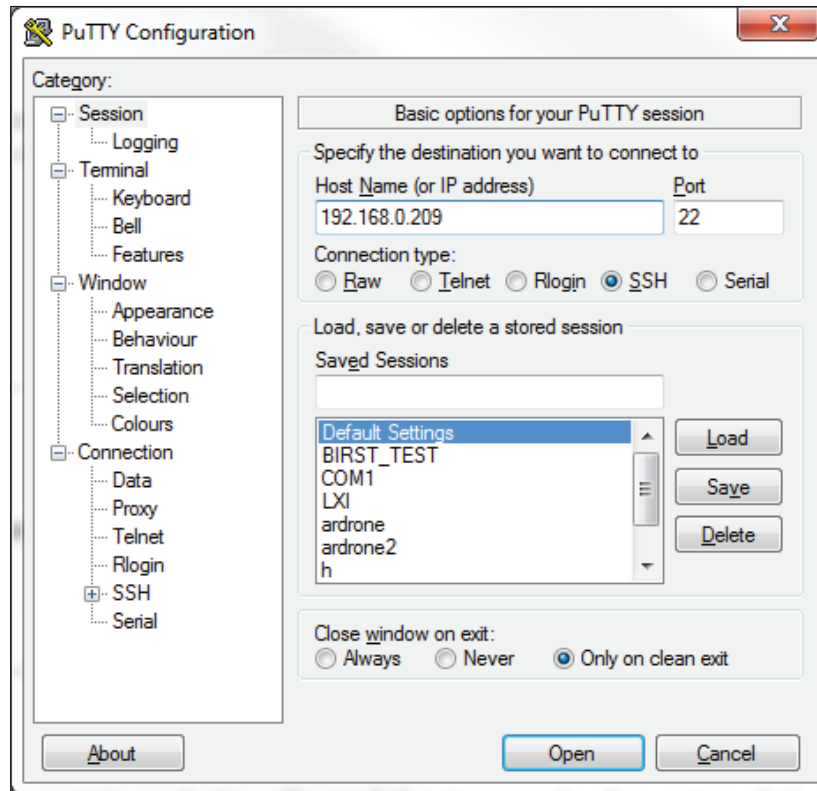
Section 5 - Programming

Instrument Control from an SSH Client

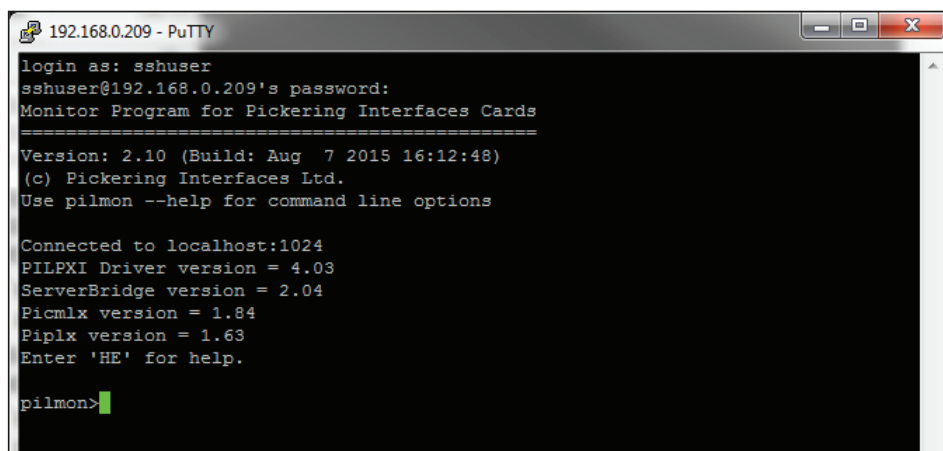
In the following examples the SSH client PuTTY is used, the user will have to adapt the instructions for any other client.

First start up the SSH client and enter the IP address of the LXI device. The user will be prompted for a login name and password.

The username for instrument control is 'sshuser', the password is that defined by the user when setting security.



Once the username and password have been supplied, the LXI device will enter an application providing a command line interface to the switch cards within the device.



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When the application is expecting input, the prompt **pilmon>** will be displayed. Card control is achieved using a set of mnemonics plus optional data. Typing **HE** in response to the prompt will list the entire command set available.

These are listed below and on the following pages.

Before attempting to control a card it is important to understand the switch architecture of that card and to know how to address the required switch. Each Pickering switch card is represented as a set of one or more sub-units, each sub-unit containing one or more switches. A sub-unit represents a collection of switches of similar type, it may represent an entire matrix, or a row of simple switches, or a power supply voltage setting.

In the case of a switch sub-unit, each switch is represented by a single bit of data and may be toggled between 0 and 1, 0 representing the off state of the switch relay and 1 representing the on state.

Table 5.1 - PILMon Command List

AttenuatorAttenuationGet	AAG	<no parameters> Get current attenuation of selected subunit. Returns float value.
AttenuatorAttenuationSet	AAS	<attenuation(float)> Set attenuation of selected subunit to <attenuation>.
AttenuatorInfo	AI	<no parameters> Get information about selected attenuator subunit. Returns values: <type>;<number of steps>;<size of each step>
AttenuatorPadValue	AP	<index(int)> Get value of attenuator pad at <index>. Returns float value.
AttenuatorType	AT	<no parameters> Get information about selected attenuator subunit in text form. Returns string in form: ATTEN(<number of steps><step size in dB>)
BatteryCurrentGet	BCG	<no parameters> Get current setting of battery simulator subunit current leak. Returns float value.
BatteryCurrentSet	BCS	<current(float)> Set battery simulator subunit current leak.
BatteryEnableGet	BEG	<no parameters> Get current setting of battery simulator subunit output states. Returns binary pattern characterizing states.
BatteryEnableSet	BES	<pattern(int)> Set battery simulator subunit output states according to pattern.
BatteryInterlockGet	BIG	<no parameters> Get interlock state of battery simulator subunit. Returns 1 if all interlocks are up or 0 if one or more interlocks are down.
BatteryVoltageGet	BVG	<no parameters> Get current setting of battery simulator subunit voltage. Returns float value.
BatteryVoltageSet	BVS	<voltage(float)> Set battery simulator subunit votage.
CalibrationReadInt	LRI	<index(int)> Read integer calibration value of pad <index>. Returns integer value.
CalibrationReadDate	LRD	<store(int)> Read calibration date of subunit. <store> can be 0-for user store 1-for factory store. Returns date in form: <year>;<day>;<interval>.
CalibrationReadFp	LRF	<store(int)> <offset(int)> Read float calibration value of <store> on position <offset>. Returns float value.
CalibrationWriteInt	LWI	<index(int)> <value(int)> Write integer calibration value <value> of pad <index>.
CalibrationWriteDate	LWD	<store(int)> <interval(int)> Write calibration date of subunit. <store> can be 0-for user store 1-for factory store. Date written is current system date <interval> is number of days for cards to be recalibrated.
CalibrationWriteFp	LWF	<store(int)> <offset(int)> <value(float)> Write float calibration value <value> of <store> on position <offset>.
CardClear	CC	<no parameters> Clear all switches on all subunits on selected card.

Section 5 - Programming

CardClearAll	CCA	<no parameters> Clear all switches on all subunits on all cards.
CardClose	CL	<no parameters> Close selected card. Card cannot be used until opened again.
CardCloseAll	CLA	<no parameters> Close all opened cards.
CardDiagnostic	CD	<no parameters> Get text diagnostic of selected card. When card is faulty diagnostics shows details about faults. Returns diagnostic text. It does not diagnose the relays, only communication.
CardId	CI	<no parameters> Get card id string in form: <card model>;<serial number>;<card revision>
CardLoc	CLO	<no parameters> Get card physical location. Returns location in form: <bus>;<device>;
CardOpen	CO	<bus(int)> <device(int)> [access(int)] Open card on physical position <bus> <device>. Optional parameter [access] modifies access setting. 1-exclusive access 0-nonexclusive access. Card is opened with exclusive access implicitly.
CardOpenAll	COA	[access(int)] Open all cards in LXI if possible. Optional parameter [access] modifies access setting. 1-exclusive access 0-nonexclusive access. Cards are opened with exclusive access implicitly.
CardStatus	CS	<no parameters> Get status of selected card. Status consists of binary flags detecting various problems of card. Result: hex integer containing status flags.
FindCountCards	FCC	<no parameters> Get number of available cards in LXI. Returns integer value.
FindFreeCards	FFC	<no parameters> Get location of all unopened cards. Returns locations of cards in form: <bus1> <device1>;<bus2> <device2>;...
InputBitRead	IBR	<bit(int)> Read state of individual <bit> of selected input subunit. Returns state of bit= 1-active 0-inactive.
InputSubunitRead	IUR	<no parameters> Read states of all bits of selected input subunit. Returns hex string representing all states of bits.
PSUEnableSet	PES	<state(int)> Set output state of PSU subunit.
PSUInfo	PI	<no parameters> Get info about selected PSU subunit. Returns info in form: <type>;<rated voltage>;<rated current>;<precision>;<capabilities(hex flags)>
PSUType	PT	<no parameters> Get info about selected PSU subunit in string form. Returns info in form: PSUDC(<rated voltage><rated current>).
PSUVoltageGet	PVG	<no parameters> Get current setting of voltage of selected PSU subunit. Returns float value.
PSUVoltageSet	PVS	<voltage(float)> Set voltage setting of selected PSU subunit.
ResistorInfo	RI	<no parameters> Get info about selected programmable precision resistor subunit. Returns: <minres>;<maxres>;<referenceres>;<precisionpc>;<precisiondelta>;<reserved>;<reserved>;<capabilities(hex flags)>
ResistorResistanceGet	RRG	<no parameters> Get output resistance of selected programmable precision resistor subunit. Returns float value.
ResistorResistanceSet	RRS	<resistance(float)> Set output resistance of selected programmable precision resistor subunit to <resistance>.
SwitchClose	SC	<bit(int)> Close switch (energize relay coil) on selected subunit of index <bit>.

Section 5 - Programming

SwitchOpen	SO	<bit(int)> Open switch (disengage relay coil) on selected subunit of index <bit>.
SwitchView	SV	<bit(int)> Get state of switch on selected subunit of index <bit>. Returns 1 if coil is energized 0 otherwise.
SwitchMask	SM	<bit(int)> Mask switch on selected subunit of index <bit>. Switch cannot be operated until unmasked.
SwitchUnmask	SU	<bit(int)> Clear mask of switch on selected subunit of index <bit>.
SwitchMaskView	SMV	<bit(int)> Get state of mask of switch on selected subunit of index <bit>. Returns 1 if mask is active 0 otherwise.
SubunitClear	UC	<no parameters> Clear all switches (set relays to their default state) on selected subunit.
SubunitClosureLimit	UCL	<no parameters> Get closure limit of selected subunit. Closure limit is maximum number of switches that can be simultaneously closed on a subunit. Returns integer value.
SubunitEnumerate	UE	<no parameters> Get number of input and output subunits on selected card. Returns: <number of input subunits>;<number of output subunits>
SubunitInfo	UI	[input(int)] Get information about subunit. Optional parameter <input> states whether to get information about input-0 or output-1 subunit. Implicitly its output. Returns information in form: <type>;<number of rows>;<number of columns>"
SubunitSettleTime	USE	<no parameters> Get time in ms that is required for relay contact to settle down after switch operation. Returns integer value.
SubunitStatus	UST	<no parameters> Get status of selected subunit. Status consists of binary flags detecting various problems of subunit. Result: hex integer containing status flags.
SubunitType	UT	[input(int)] Get string info about selected subunit. Return string contains description of the subunit. Optional parameter <input> states whether to get information about input-0 or output-1 subunit. Implicitly its output. For example: MATRIX(<columns>X<rows>)"
SubunitMaskClear	UMC	<no parameters> Clears all mask bits on selected subunit.
SubunitMaskView	UMV	<no parameters> View states of all mask bits on selected subunit. Returns hex string describing mask. Each bit in hex string represents 1 bit of mask.
SubunitMaskWrite	UMW	<pattern(hex string)> Writes hex string converted to bit array into subunit mask bits. Note that hex string must be started with \0x\"notation".
SubunitView	UV	<no parameters> View states of all switches on selected subunit. Returns hex string describing states of switches. Each bit in hex string represents 1 switch.
SubunitWrite	UW	<pattern(hex string)> Writes hex string converted to bit array into subunit. Note that hex string must be started with \0x\"notation".
VsourceEnableGet	VEG	<no parameters> Get enabled states of selected voltage source subunit as pattern. Returns integer pattern where each bit represents one voltage source output.
VsourceEnableSet	VES	<pattern(int)> Set enabled states of selected voltage source subunit. Each bit in <pattern> represents one voltage source output.
VsourceVoltageGet	VVG	<no parameters> Get voltage settings of selected voltage source subunit. Return float value.
VsourceVoltageSet	VVS	<voltage(float)> Set <voltage> of selected voltage source subunit.

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XpointClose	XC	<column(int)> <row(int)> Close switch (energize relay coil) on selected matrix subunit on position <column> <row>.
XpointOpen	XO	<column(int)> <row(int)> Open switch (disengage relay coil) on selected matrix subunit in position <column> <row>.
XpointView	XV	<column(int)> <row(int)> Get state of switch on selected matrix subunit in position <column> <row>. Returns 1 if coil is energized 0 otherwise."
XpointMask	XM	<column(int)> <row(int)> Mask switch on selected matrix subunit in position <column> <row>. Switch can't be operated until unmasked."
XpointUnmask	XU	<column(int)> <row(int)> Clear mask of switch on selected matrix subunit in position <column> <row>.
XpointMaskView	XMV	<column(int)> <row(int)> Get state of mask of switch on selected matrix subunit in position <column> <row>. Returns 1 if mask is active 0 otherwise.
ModeSet	MS	Set driver in different working mode. This feature is to be used only by experts!
MultiAccessSet	MAS	<access(int)> [card(int)] Modify access rights for selected card. <access> is 1 for exclusive access 0 for multiaccess. Optional parameter [card] can be used to change access rights for unopened card.
MultiAccessGet	MAG	[card(int)] Query access rights for selected card. Optional parameter [card] can be used to query access rights for unopened card. Return string describing access rights.
SubunitSelect	USL	<card(int)> <subunit(int)> [input(int)] Selects active <subunit> on card number <card>. Optional parameter <input> states whether to get information about input-0 or output-1 subunit. Implicitly it's output. When called without any parameters returns actual settings.
List	LS	List all cards and show their location. For opened cards more details will be displayed.
Help	HE	Show help about all commands. Help <command> shows more detailed explanation of syntax and meaning of commands.
Quit	Q	Ends work with pilmon. All resources will be freed but states of cards remain unchanged.
SetAttribute	SA	<attrCode(uint)> <attrValue(uint, double)> Set the value of the specific Attribute (defined by attribute code (<attrCode>)). See Table 5.2 e.g. SA 0x40A 1 (Set Low range on Thermocouple card)
GetAttribute	GA	Get the value of the specific Attribute (defined by attribute code (<attrCode>)). Returns attribute value: <attrValue>. See Table 5.2 e.g. GA 0x40A (returns the set value)

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Table 5.2 - List of Attribute Codes

ATTR_MODE	0x401	Gets/Sets DWORD attribute value of Mode of the Card. Applicable for Fault Insertion 40-200 and 40-201 cards to add/remove Fault bus restrictions. Applicable Parameters: UNRESTRICTED = 0 RESTRICTED = 1
ATTR_TS_SET_RANGE	0x40A	Gets/Sets different range on the Thermocouple cards. Applicable 41-760 and 41 761 products. Applicable Parameters: TS_AUTO_RANGE = 0 (Default) TS_LOW_RANGE = 1 TS_MED_RANGE = 2 TS_HIGH_RANGE = 3

So a quick example to operate a bit on the card would be:

COA > open all cards

LS > list the cards

USL 1 1 > Select the card 1 and sub-unit 1

SC 1 > energize bit 1

```

ServerBridge version = 2.02
Picmlx version = 1.83
Piplx version = 1.61
Enter 'HE' for help.

pilmon>ls
Card 1:
  Location = bus 4, slot 0
  ID string = 40-132-102,290096,2.00
  Input sub-units: none
  Output sub-units: 1 = SWITCH(16)
  Status = 0x0 (No errors.)
Card 2:
  Location = bus 4, slot 1
  ID string = 40-670-022,231003,1.00
  Input sub-units: none
  Output sub-units: 1 = MUXM(10)
  Status = 0x0 (No errors.)
OK.
pilmon>usl 1 1
OK.
pilmon>sc 1
OK.
pilmon>

```

Section 6 - Troubleshooting

Refer to the Warnings and Cautions at the front of this manual.



LXI Status Indicators

Status Indicator / Colour				Status Description
PWR	RDY	ERR	LAN	
 Green	 Off	 Off	 Off	Powering Up Status: This state occurs when the LXI unit was turned off and the user turns it on. This state will exist for ~10-15 seconds.
 Green	 Orange	 Orange	 Orange	PI Services Init. Status: This state triggers after powering up status and it means that PI services are initialising (pilserv is starting and initialising). This state should last <1 second.
 Green	 Green	 Off	 Green	Normal Operation Status: LXI unit is turned on, Ethernet connection is established (LXI unit has assigned IP address from DHCP/AutoIP or static range), ServerBridge is running and listening for incoming connections.
 Green	 Green	 Off	 Red	Network Failure Status: LXI unit is turned on, but the Ethernet connection is not established (LXI unit is not connected to any network and it has not assigned an IP address). Client cannot access the unit or control any of the plugged in cards.
 Green	 Off	 Red	 Green	ServerBridge Failure Status: LXI unit is turned on, Ethernet connection is established (LXI unit has assigned IP address from DHCP/AutoIP or static range), ServerBridge is not running. Client can access the LXI unit thru the web interface, but cannot control any of the plugged in cards.
 Green	 Off/Green (toggling between off & green)	 Off/Red (toggling between off & red)	 Green	ServerBridge Failure Status: LXI unit is turned on, Ethernet connection is established (LXI unit has assigned IP address from DHCP/AutoIP or static range), ServerBridge is still restarting. Client can access the LXI unit thru the web interface, but cannot control any of the plugged in cards. The cause of the fault is usually the pilpxi driver, either of the third party drivers: pi743/pxm78xx or a bug in ServerBridge.
 Green	 Green	 Off	 Orange	Ethernet Change Notification Status: LXI unit is turned on, Ethernet connection is changing. This is usually caused by user action: connected ethernet cable or LAN configuration change. ServerBridge is terminated in this state and waiting to establish the Ethernet connection.

Section 6 - Troubleshooting

- Q** I have switched my chassis on, but why is the PWR light not illuminated?
- A** No PWR light could mean one of the following issues:
1. Power supply not connected to the chassis, mains lead not connected to the power supply, mains supply off or power supply faulty.
 2. A faulty PXI module has caused an overcurrent in which case the internal PSU will be disabled - switch the chassis off, remove the PXI modules and switch on again to see if PWR is illuminated.
 3. There is a hardware fault - contact your local sales office
- Q** The 100 or 1000 light is on and the PWR light is on, but why does the LAN light flash green?
- A** This is normal operation, it is used as visual identification to locate the chassis in a densely populated test rack environment.
- Q** The 100 or 1000 light is on and the PWR light is on, but why does the LAN light flash red?
- A** There is an IP conflict (you have probably configured your chassis to a static IP address which is in collision with another).
- Q** The 100 or 1000 light is on and the PWR light is on, but why is the LAN light illuminated red?
- A** The steady red LAN light indication means one of the following issues:
1. The Ethernet cable is disconnected.
 2. Failure to renew an already acquired DHCP lease (in this case the chassis will revert to AutoIP configuration and will look for new renewal value).
 3. Detection of a duplicate IP address - (you have probably configured your chassis to a static IP address which is in collision with another).
 4. Failure to acquire a valid IP address
- Q** Why do I have only a green PWR light?
- A** The chassis is either in USB mode or there is a hardware fault with your chassis - contact your local sales office.
- Q** What does the orange LAN light indication mean? (The 100 or 1000 light is on and the PWR light is illuminated green).
- A** This means that the chassis is changing the LAN configuration.
- Q** How do I know when I can begin communicating with my chassis?
- A** When the 100 or 1000 light is on, the PWR light on and the LAN light illuminated green.

Please contact your local Pickering Interfaces sales office for any further support or email: support@pickeringtest.com

You can also visit our website for product updates, help and downloadable materials at: pickeringtest.com

Section 6 - Troubleshooting

Unit Reset

If the unit's operation is inconsistent, you can try cycling the power using the power switch located on the rear panel to re-boot the chassis. Alternatively, you can press the recessed "LAN Reset" button on the rear panel of the chassis causing it to be set back to factory default conditions. This will destroy all user configuration and reset the LAN configuration (enabled DHCP and AutoIP).

Part Replacement

For replacement and service of internal parts, please contact your Pickering representative. Failure to do so could affect your warranty.

Battery Location

The 60-105 Chassis has an internally mounted back-up battery located on the controller board as shown in the figure below. The battery type is a CR1616 3 Volt lithium coin cell.

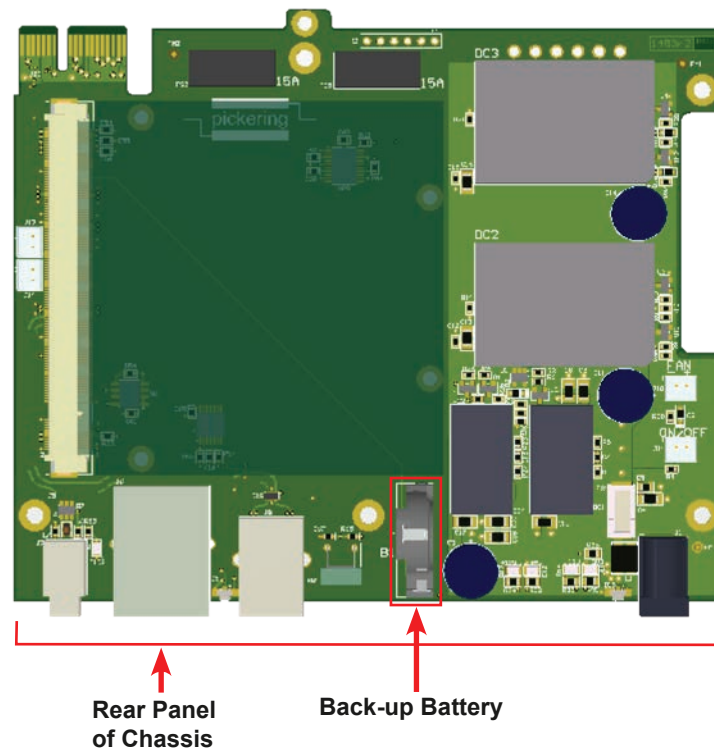


Figure 6.1 - Location of Back-up Battery on the 60-105 Controller PCB

Appendix A - Chassis Mounting Kits

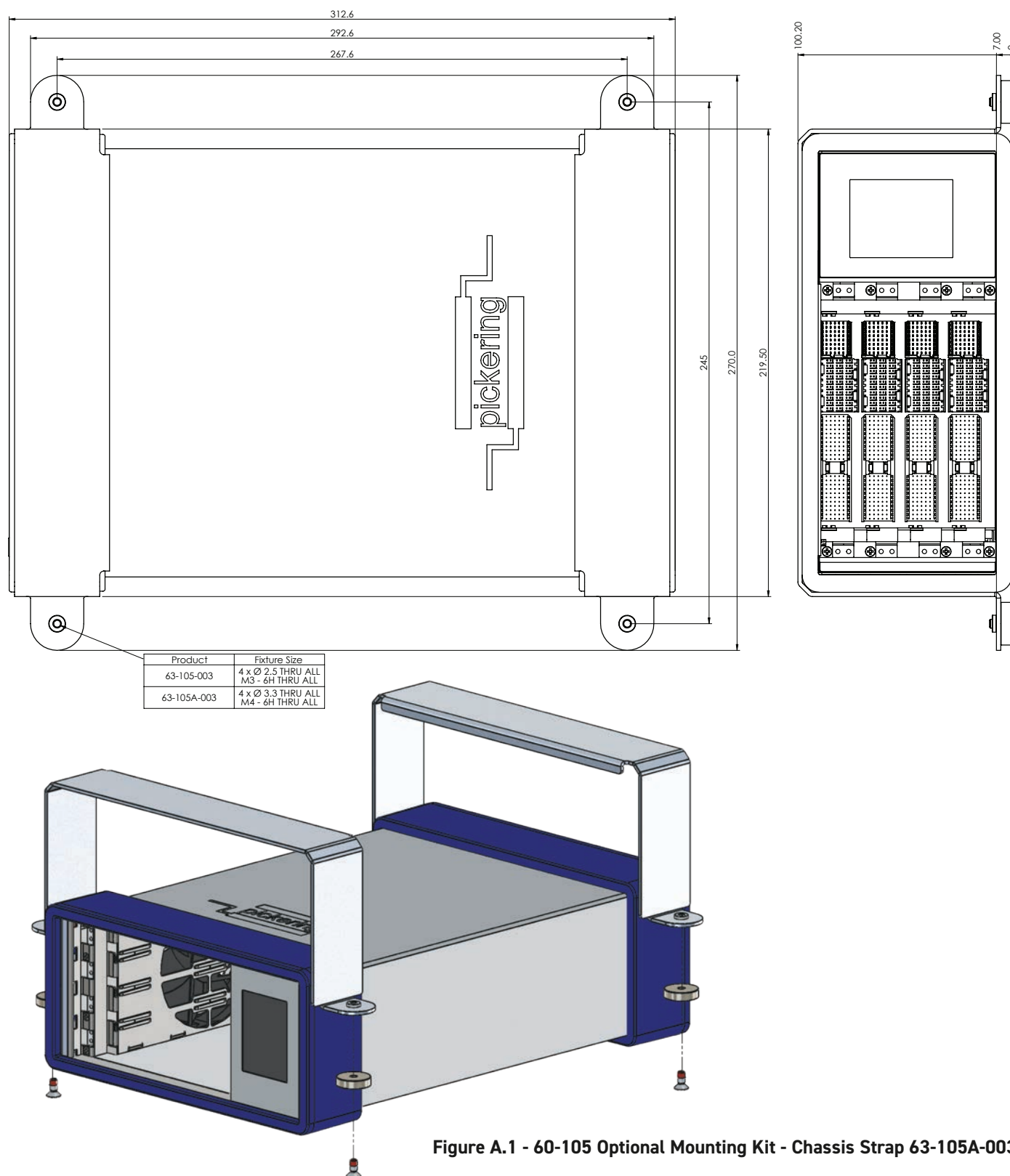


Figure A.1 - 60-105 Optional Mounting Kit - Chassis Strap 63-105A-003

Note: The 63-105A-003 is supplied primarily as a magnetic fixing method. If required, the magnets can be removed and mechanical fixings used.

Appendix A - Chassis Mounting Kits

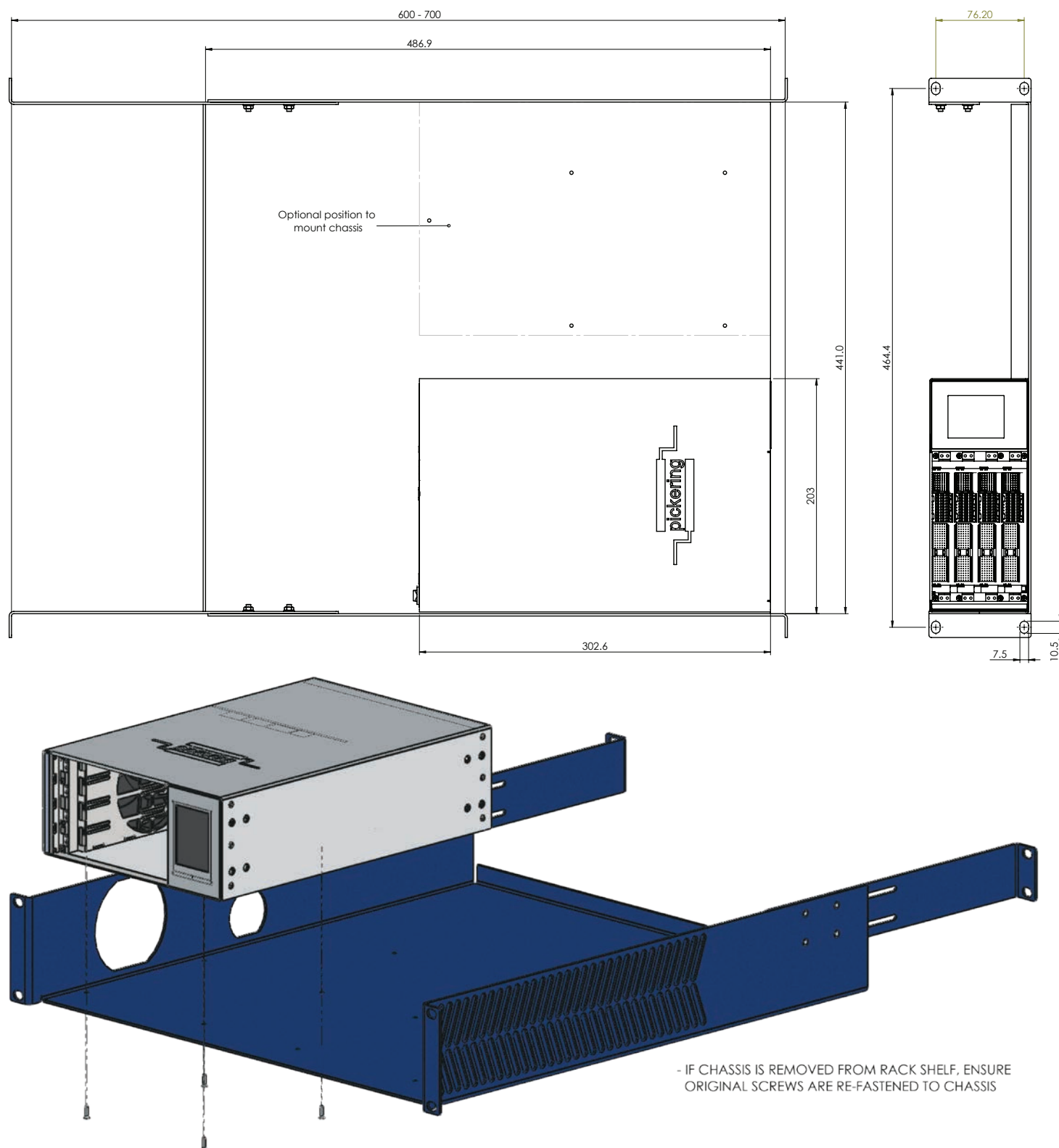


Figure A.2 - 60-105 Optional Mounting Kit - Rack Shelf Mounting 63-105-004
(this option requires the removal of the chassis' rubber bump strips)

Appendix A - Chassis Mounting Kits

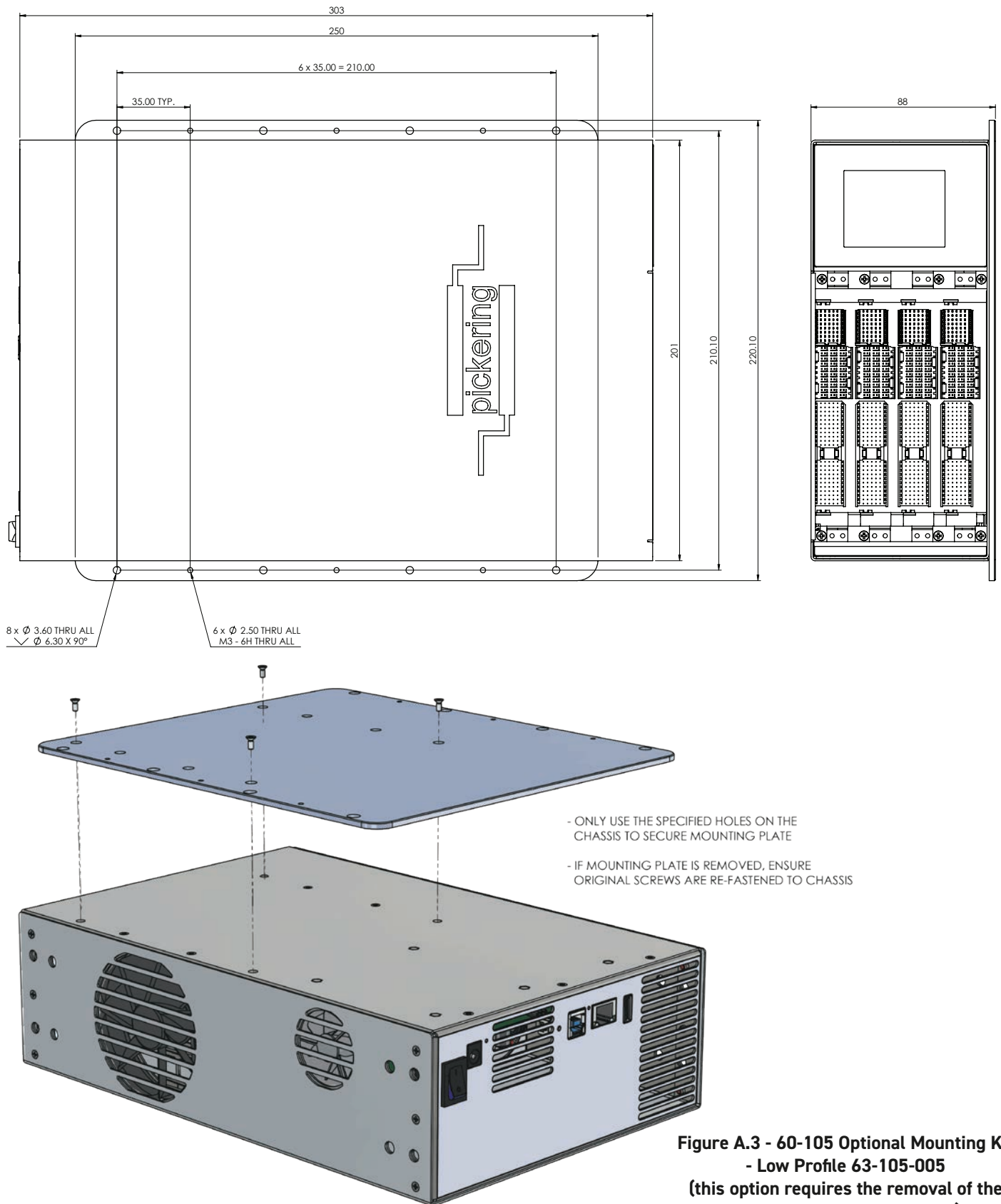


Figure A.3 - 60-105 Optional Mounting Kit
- Low Profile 63-105-005
 (this option requires the removal of the chassis' rubber bump strips)